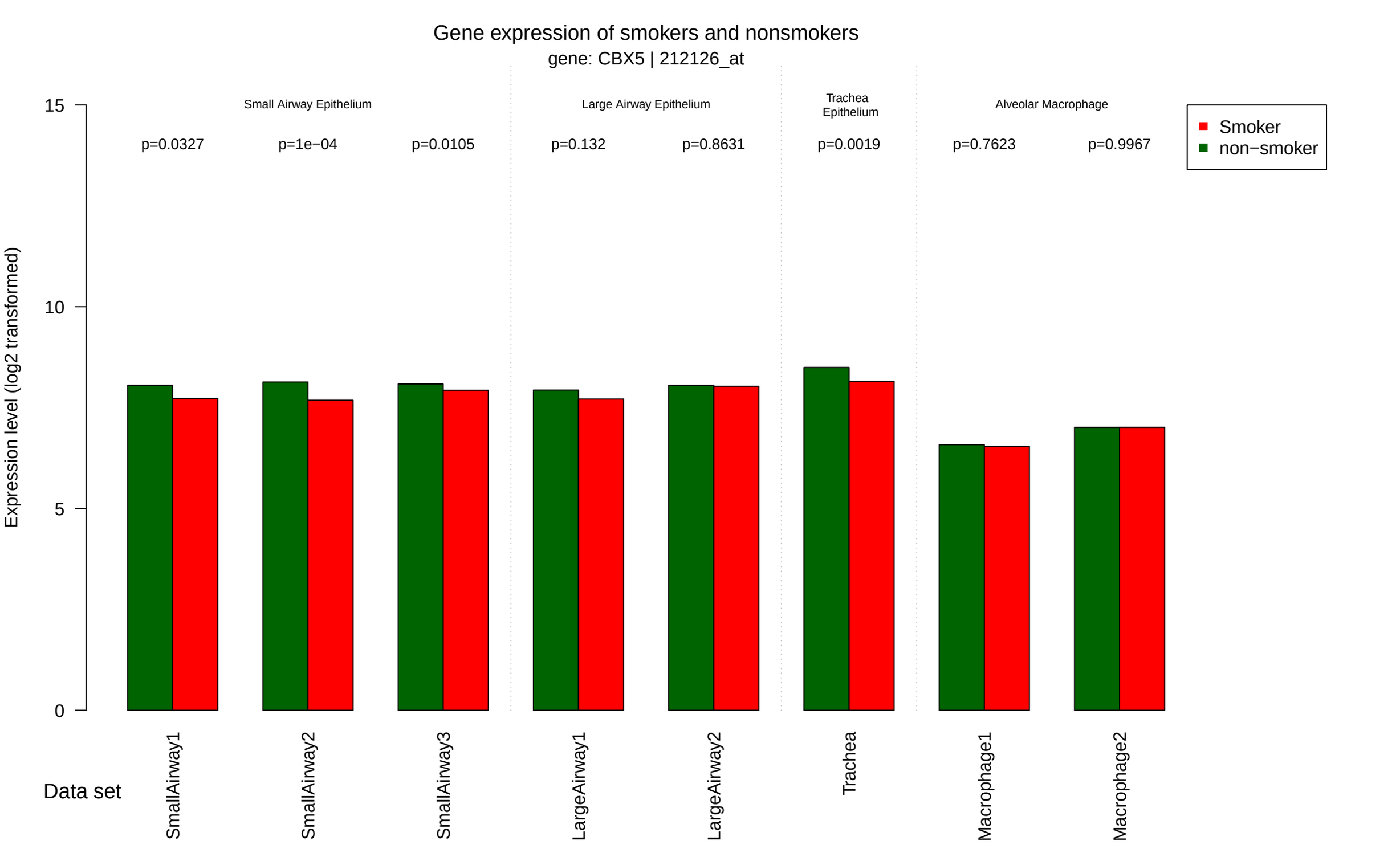
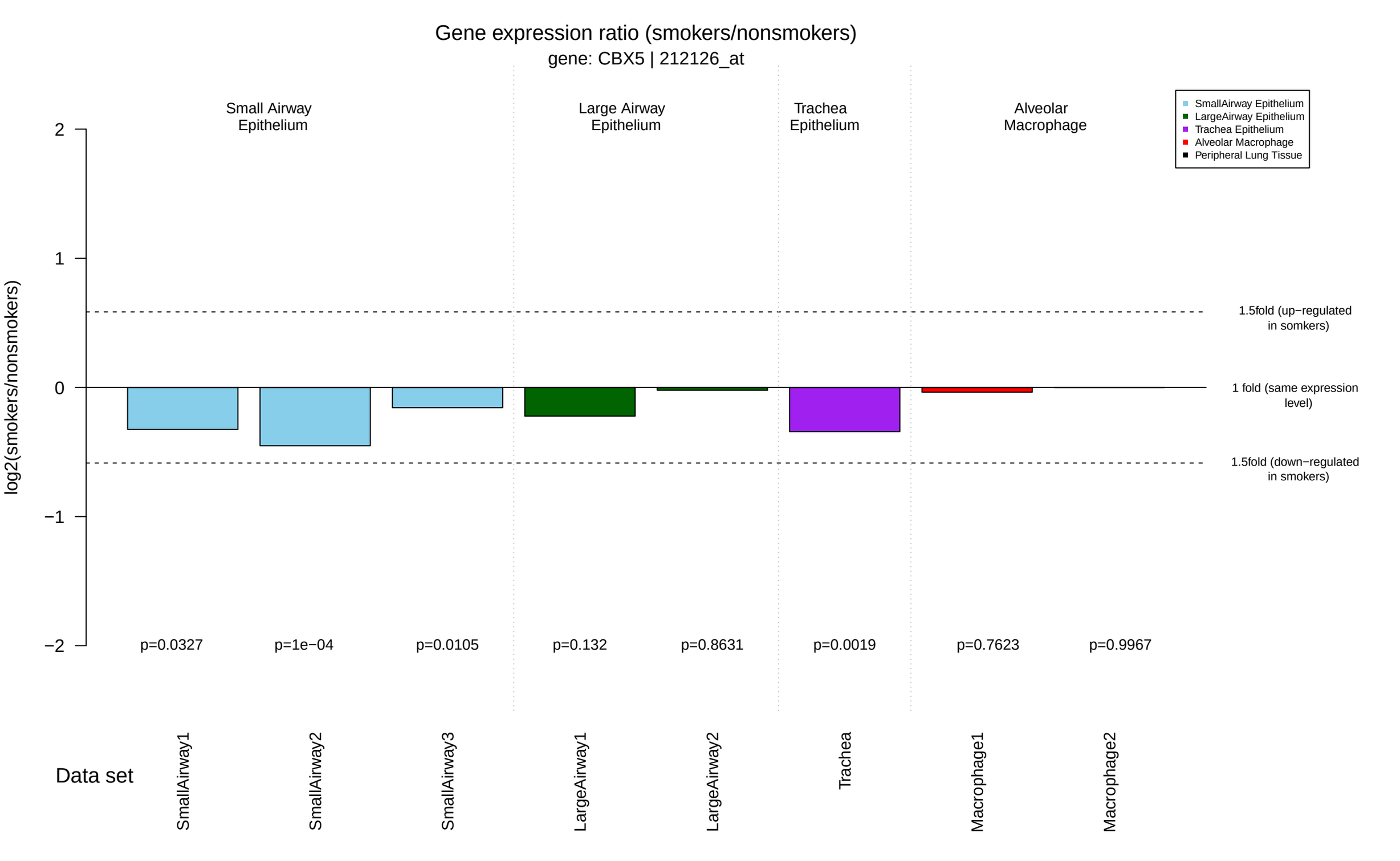
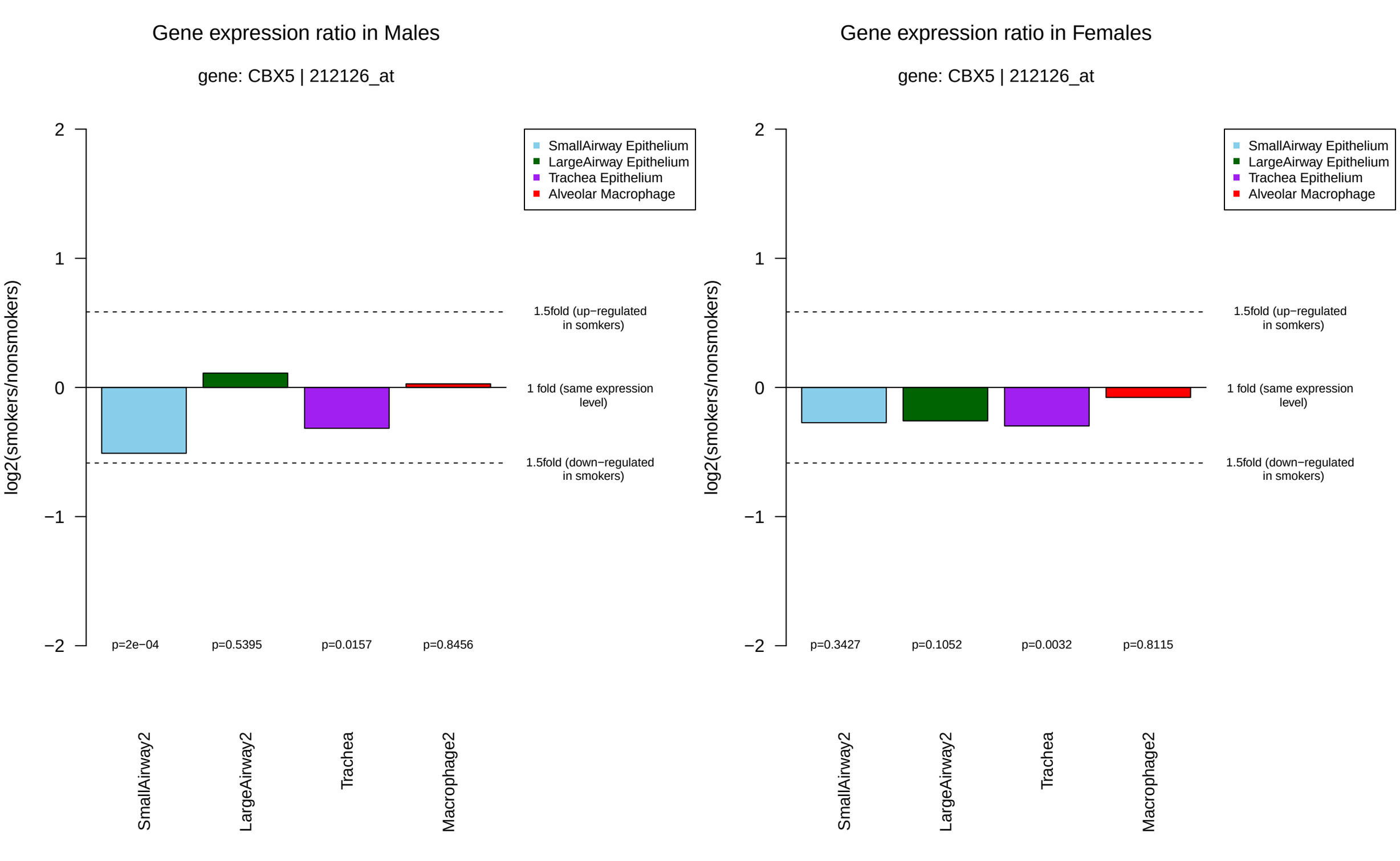
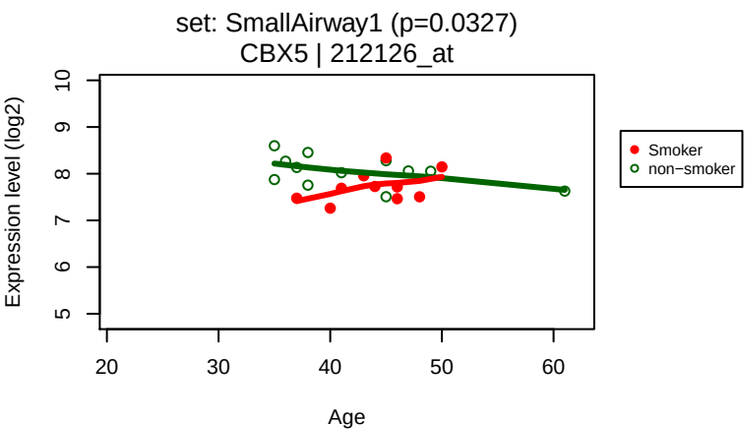
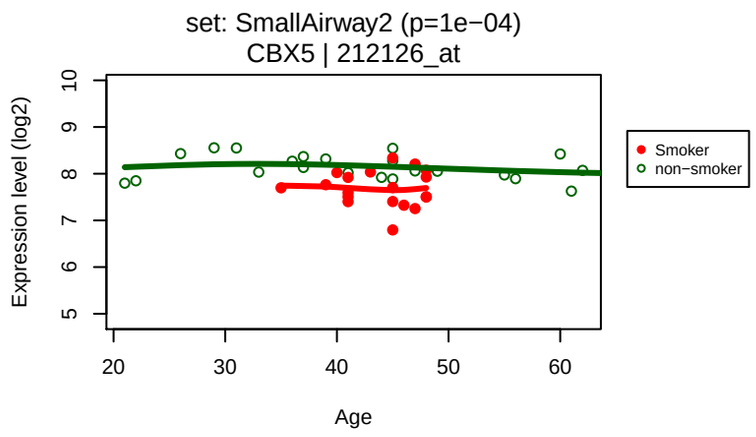
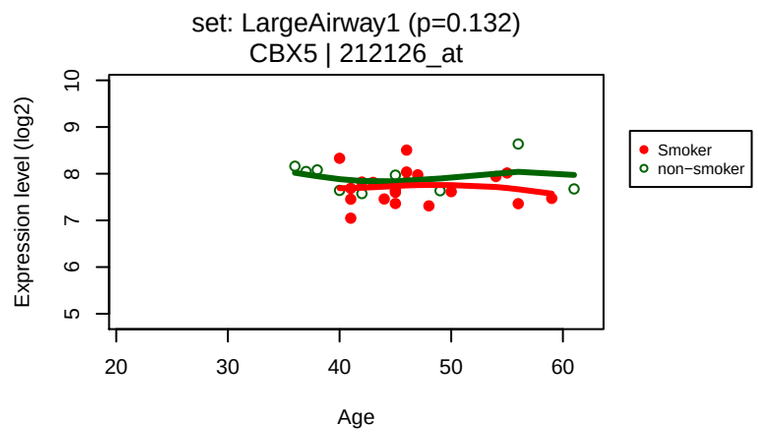

Gene expression of smokers and nonsmokers
gene: CBX5 | 212126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0327
p=1e−04
p=0.0105
p=0.132
p=0.8631
p=0.0019
p=0.7623
p=0.9967
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

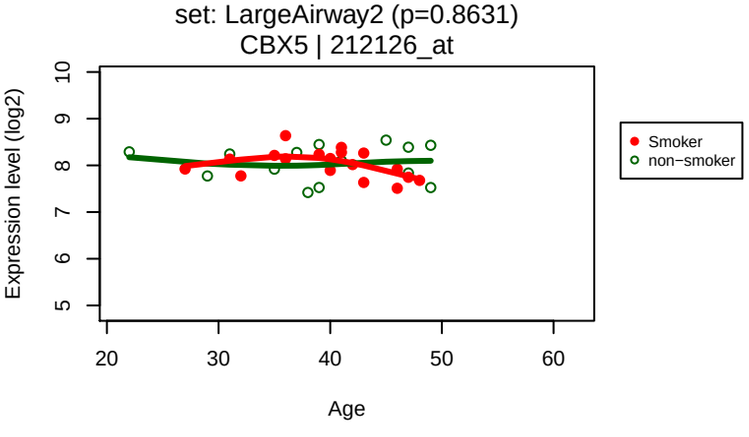

Gene expression ratio (smokers/nonsmokers)
gene: CBX5 | 212126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0327
p=1e−04
p=0.0105
p=0.132
p=0.8631
p=0.0019
p=0.7623
p=0.9967
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

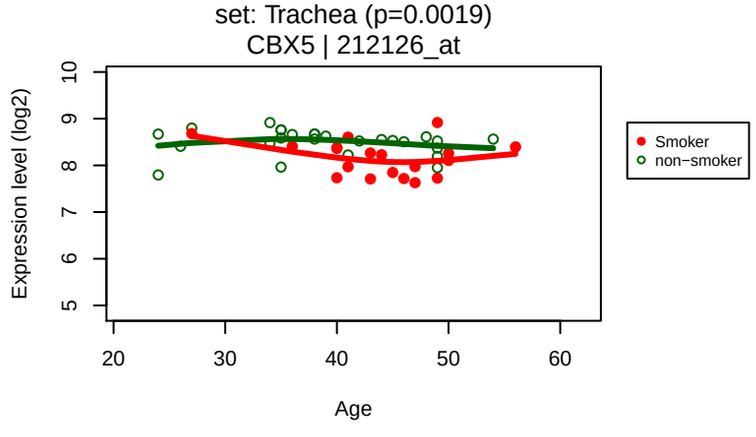

Gene expression ratio in Males
gene: CBX5 | 212126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=2e−04
p=0.5395
p=0.0157
p=0.8456
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CBX5 | 212126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3427
p=0.1052
p=0.0032
p=0.8115
SmallAirway2
LargeAirway2
Trachea
Macrophage2

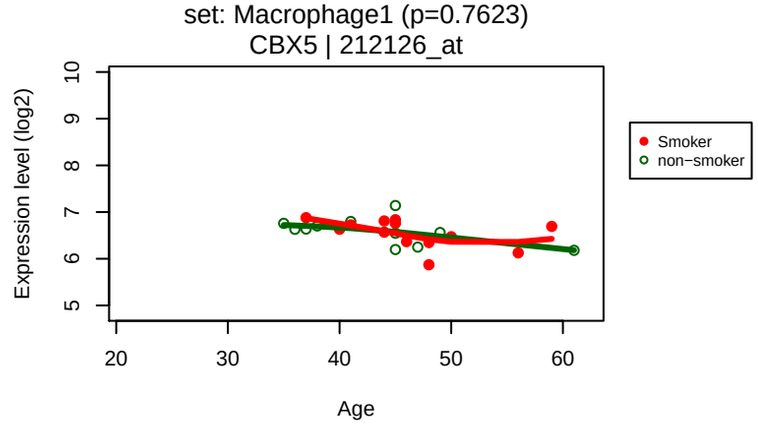

set: SmallAirway1 (p=0.0327)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker

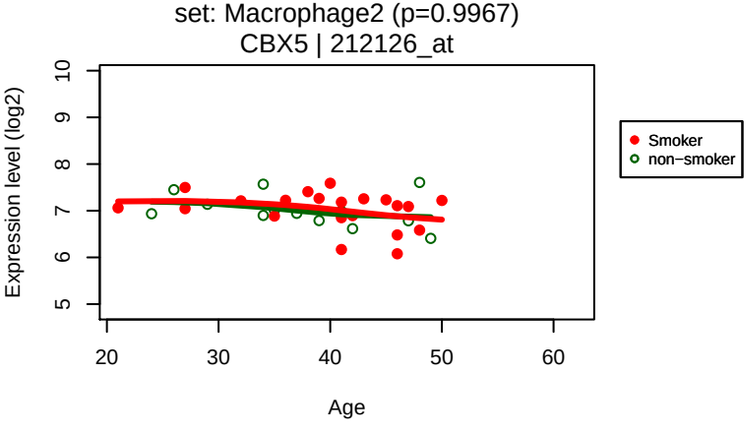

set: SmallAirway2 (p=1e−04)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker

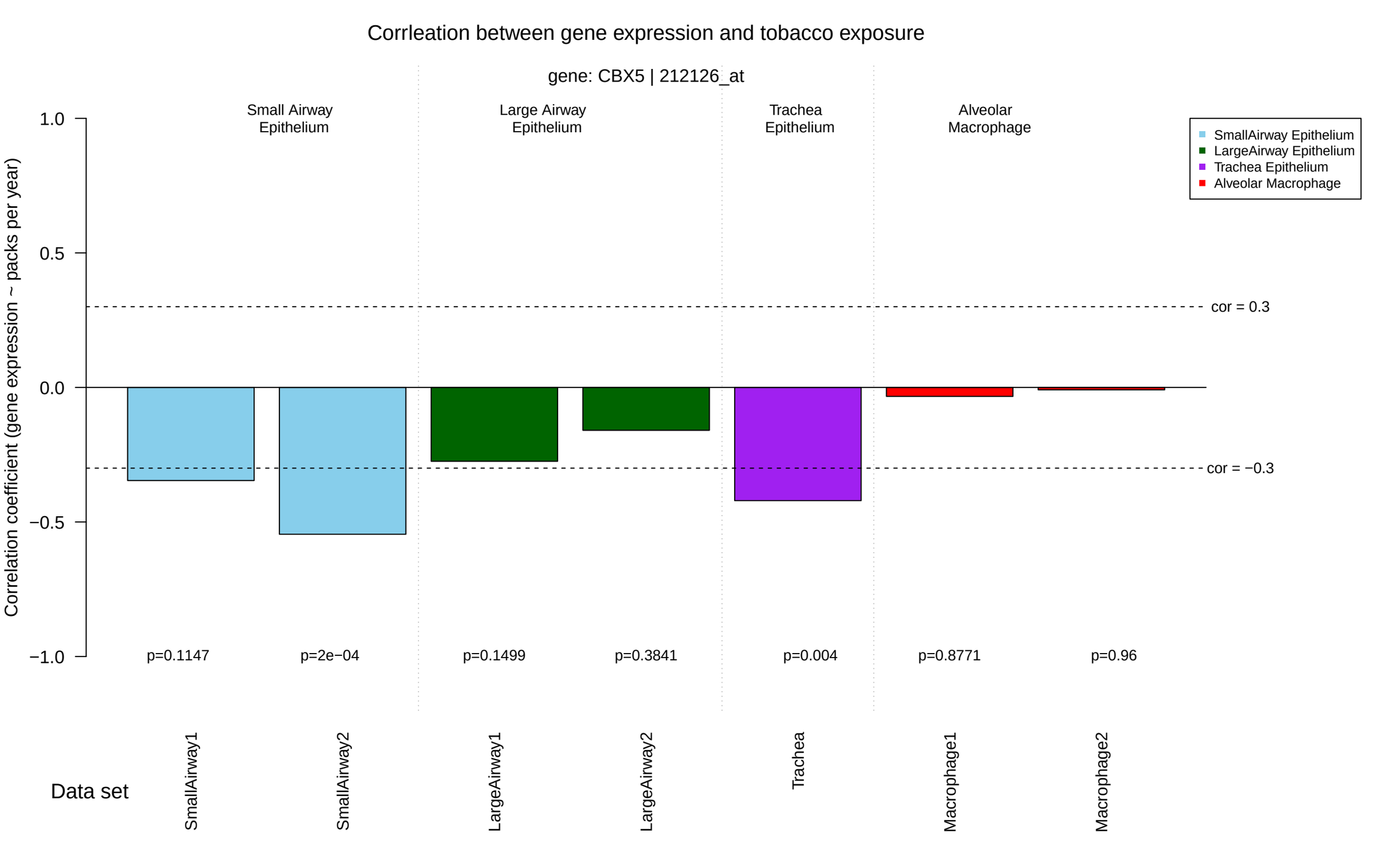

set: LargeAirway1 (p=0.132)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8631)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0019)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7623)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9967)
CBX5 | 212126_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CBX5 | 212126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1147
p=2e−04
p=0.1499
p=0.3841
p=0.004
p=0.8771
p=0.96
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2